Question 1 : Explain the difference between Absolute, Relative, and Mixed Cell Referencing in Excel with examples.
Answer :
[11:41 am, 07/04/2026] Minakshi Janbandhu: 1. Absolute Cell Reference
- Definition: The cell reference remains fixed, no matter where the formula is copied.
- Notation: Uses a dollar sign $ before the column letter and row number.
- Example:
- Formula: =A1*B2
- If copied to another cell, B2 will always point to cell B2.
Use Case: When you want to multiply values by a constant (e.g., tax rate, interest rate).

📌 2. Relative Cell Reference
- Definition: The cell reference changes relative to the position where the formula is copied.
- Notation: No dollar signs.
- Example:
- Formula: =A1*B1
- If copied one row down, it becomes =A2*B2.
Use Case: When applying the same calculation across multiple rows or columns.

📌 3. Mixed Cell Reference
- Definition: Part of the reference is fixed (absolute), and part is relative.
- Notation: Dollar sign before either the column or the row.
- Examples:
- =$A1 → Column A is fixed, row changes when copied.
- =A$1 → Row 1 is fixed, column changes when copied.
Use Case: Useful in tables where one dimension (row or column) must stay constant while the other adjusts

Question 2 : What is a Macro in Excel? How does it help in automation?
Answer :

What is a Macro in Excel?

- A Macro is a sequence of instructions that automates repetitive tasks in Excel.
- Technically, it is a small program written in VBA (Visual Basic for Applications) that records and executes actions like formatting, calculations, or data manipulation.
- Instead of manually performing the same steps repeatedly, you can record them once as a macro and run it whenever needed.

📌 How Macros Help in Automation

1. Save Time
    - Automates repetitive tasks such as formatting reports, applying formulas, or importing data.

    - Example: Instead of manually applying bold, color, and borders to a table every week, a macro can do it instantly.
2. Reduce Errors
    - Ensures consistency by performing tasks exactly the same way each time.
    - Example: A macro can standardize how monthly sales data is cleaned and summarized.
3. Increase Productivity
    - Allows users to focus on analysis rather than manual work.
    - Example: Automating the generation of pivot tables and charts for management reports.
4. Complex Task Execution

- Can handle multi-step processes that would be tedious manually.
- Example: Importing data from multiple sheets, cleaning it, and creating a consolidated

Question 3 : What are Text Functions in Excel? Mention any five with examples.

Answer :

What are Text Functions in Excel?

- Text Functions are built-in formulas that allow you to manipulate, format, and analyze text strings in Excel.
- They are especially useful for cleaning data, extracting information, and preparing reports.

📌 Five Common Text Functions with Examples

1. LEFT()

- Purpose: Extracts a specified number of characters from the left side of a text string.
- Example: `=LEFT("Excel", 3)` → Result: "Exc"

1. RIGHT()

- Purpose: Extracts a specified number of characters from the right side of a text string.
- Example: `=RIGHT("Excel", 2)` → Result: "el"

1. LEN()

- Purpose: Returns the number of characters in a text string (including spaces).
- Example: `=LEN("Excel 2026")` → Result: 10

1. CONCATENATE() / CONCAT()

- Purpose: Joins two or more text strings into one.
- Example: `=CONCAT("Hello", " ", "World")` → Result: "Hello World"

  (Note: In newer versions of Excel, `CONCAT()` replaces `CONCATENATE()`.)

1. TRIM()

- Purpose: Removes extra spaces from text, leaving only single spaces between words.
- Example: `=TRIM(" Excel Functions ")` → Result: "Excel Functions"

Question 4 : What is the use of Scenario Manager in decision making?

Answer :

What is Scenario Manager?

- **Scenario Manager** is a tool in Excel (part of the What-If Analysis feature) that allows you to create and compare different sets of input values (scenarios) to see how they affect the outcome of formulas.
- It's especially useful when you want to test multiple possibilities without manually changing data each time.

Use in Decision Making

1. **Compare Alternatives**
    - Helps managers evaluate different business strategies by changing assumptions.
    - Example: Compare profit outcomes under "Best Case," "Worst Case," and "Most Likely" scenarios.
2. **Forecasting & Planning**
    - Useful for financial planning, budgeting, and sales forecasting.
    - Example: Test how revenue changes if sales increase by 10%, 20%, or 30%.
3. **Risk Analysis**
    - Allows decision makers to see the impact of uncertain variables.
    - Example: Assess how changes in raw material costs affect overall profit margins.
4. **Quick Insights**

- Instead of recalculating manually, Scenario Manager instantly shows results for each scenario side by side.
- Example: A company can quickly decide whether to expand production based on cost and demand scenarios.

Question 5 : Define the purpose of VLOOKUP and HLOOKUP. How are they different from XLOOKUP? Which among XLOOKUP and INDEX-MATCH is best while usage?

Answer :

Here's a detailed explanation of **VLOOKUP, HLOOKUP, XLOOKUP, and INDEX-MATCH** with their differences and best usage:

VLOOKUP (Vertical Lookup)

- **Purpose**: Searches for a value in the first column of a table and returns a value from another column in the same row.
- **Syntax**: `=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

- **Example**: `=VLOOKUP(101, A2:D10, 3, FALSE)` → Finds product ID **101** in column A and returns the value from the 3rd column.

HLOOKUP (Horizontal Lookup)

- **Purpose**: Similar to VLOOKUP, but searches for a value in the first row of a table and returns a value from another row in the same column.
- **Syntax**: `=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])`
- **Example**: `=HLOOKUP("Price", A1:H3, 2, FALSE)` → Finds the header **"Price"** in row 1 and returns the value from row 2.

XLOOKUP (Modern Replacement)

- **Purpose**: A more powerful and flexible lookup function introduced in newer versions of Excel.
- **Syntax**: `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])`
- **Advantages over VLOOKUP/HLOOKUP**:
- Can look **both vertically and horizontally**.
- No need to count column/row numbers — directly specify return array.
- Handles **exact matches by default**.
- Can return multiple values (arrays).
- Allows **custom error messages** if value not found.

Example:
`=XLOOKUP(101, A2:A10, C2:C10, "Not Found")` → Finds product ID **101** in column A and returns the corresponding value from column C.

INDEX-MATCH Combination

INDEX: Returns the value of a cell at a given row and column in a range.

- **MATCH**: Returns the position of a value in a range.
- **Combined Use**: `=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`
- **Advantages**:
- More flexible than VLOOKUP/HLOOKUP (can look left or right).
- Often faster in large datasets.
- Widely used before XLOOKUP was introduced.

Example:
`=INDEX(C2:C10, MATCH(101, A2:A10, 0))` → Finds product ID **101** in column A and returns the corresponding value from column C.

XLOOKUP vs INDEX-MATCH — Which is Best?

- **XLOOKUP**:
  - Easier to write and understand.
  - More powerful (handles errors, multiple results, flexible search modes).
  - Recommended for modern Excel users.
- **INDEX-MATCH**:
- Still useful for compatibility with older Excel versions (pre-Office 365).
- Preferred by advanced users who need fine-grained control in legacy environments.

Question 6 : Create a dataset of 8 employees with joining dates. Use the dummy dataset to calculate their experience in years and months.

Answer :

Here's a dummy dataset of 8 employees with joining dates, along with a calculation of their experience in years and months (assuming today's date is 07-Apr-2026):

EmployeeID	EmployeeName	JoiningDate	Experience (Years-Months)
E001	Rahul Sharma	15-Jan-2020	6 Years 2 Months
E002	Priya Mehta	10-Jul-2018	7 Years 9 Months
E003	Amit Verma	25-Mar-2022	4 Years 0 Months
E004	Sneha Patil	01-Dec-2015	10 Years 4 Months
E0015	Arjun Singh	18-Sep-2021	4 Years 7 Months
E006	Neha Gupta	05-Feb-2019	7 Years 2 Months
E007	Karan Joshi	30-Oct-2023	2 Years 5 Months
E008	Meena Rao	12-Aug-2017	8 Years 8 Months

Formula to Calculate Experience

In Excel, you can calculate Years and Months of experience using:

=DATEDIF(JoiningDate, TODAY(), "Y") & " Years " & DATEDIF(JoiningDate, TODAY(), "YM") & " Months"

Question 7 : You are provided with a dataset containing details of 100 road accidents.Design a Road Accident Analysis Dashboard using Pivot Tables and Slicers in Excel.

Answer :

1. Data Preparation

Ensure the dataset contains relevant fields such as:

Date of Accident

Location

Vehicle Type

Cause of Accident

Severity (Minor, Major, Fatal)

Time of Day

Number of Casualties

Clean the data (remove blanks, ensure consistent categories).

2. Create Pivot Tables

Generate multiple Pivot Tables to summarize accident data:

Accidents by Location → Count of accidents per location.

Accidents by Vehicle Type → Count of accidents per vehicle type.

Accidents by Cause → Count of accidents per cause.

Accidents by Severity → Distribution of minor, major, and fatal accidents.

Accidents by Time of Day → Trend of accidents across morning, afternoon, evening, night.

3. Add Pivot Charts

Convert Pivot Tables into charts for visualization:

Bar Chart → Accidents by Location.

Pie Chart → Vehicle Type distribution.

Column Chart → Causes of accidents.

Stacked Column Chart → Severity distribution.

Line Chart → Accident trend over time.

4. Insert Slicers

Add slicers for interactivity:

Location, Vehicle Type, Severity, Date.

Slicers allow filtering across all Pivot Tables simultaneously, making the dashboard dynamic.

5. Dashboard Layout

Place all charts on a single worksheet.

Position slicers at the top or side for easy filtering.

Use consistent colors (e.g., red for fatal accidents, yellow for minor).

Add a clear title: “Road Accident Analysis Dashboard”.

6. Insights from Dashboard

Identify accident-prone locations.

Determine which vehicle types are most involved.

Highlight leading causes (e.g., speeding, drunk driving).

Show severity distribution (minor vs. fatal).

Detect peak accident times (rush hours, late night).

Final Deliverable

The Road Accident Analysis Dashboard provides stakeholders (traffic police, city planners, safety officers) with an interactive tool to:

Filter accident data by multiple dimensions.

Visualize accident hotspots and patterns.

Make data-driven decisions for road safety improvements.

Question 8 : Create a table of 10 products with stock levels. Use Macros to highlight: ● Low stock (less than 10 units) in red ● Overstock (more than 50 units) in green.

Answer :

Stock Level Highlighting with Macros

We create a table of 10 products with their stock levels, then use VBA Macros to automatically highlight low stock (<10 units) in red and overstock (>50 units) in green.

Product ID	Product Name	Stock Level
P001	Laptop	8
P002	Smartphone	12
P003	Tablet	55
P004	Headphones	6
P005	Keyboard	25
P006	Mouse	52
P007	Monitor	9
P008	Printer	15

P009	**Router**	**4**
P010	**Smartwatch**	**60**

```
Sub HighlightStockLevels()
    Dim ws As Worksheet
    Dim rng As Range
    Dim cell As Range

    'Set worksheet and range
    Set ws = ThisWorkbook.Sheets("Sheet1")
    Set rng = ws.Range("C2:C11") 'Stock levels column

    'Loop through each cell
    For Each cell In rng
        If IsNumeric(cell.Value) Then
            If cell.Value < 10 Then
                cell.Interior.Color = vbRed
            ElseIf cell.Value > 50 Then
                cell.Interior.Color = vbGreen
            Else
                cell.Interior.Color = xlNone
            End If
        End If
```

```
    Next cell

End Sub
```

3. Steps to Implement

Enter the product table in Excel (Products in column B, Stock Levels in column C).

Press Alt + F11 to open the VBA editor.

Insert a new module and paste the above code.

Close the editor and run the macro (Alt + F8 → HighlightStockLevels → Run).

The macro will automatically highlight:

Red → Stock less than 10 units.

Green → Stock more than 50 units.

4. Outcome

Products with dangerously low stock (e.g., Laptop, Headphones, Router) are highlighted in red.

Products with excess stock (e.g., Tablet, Mouse, Smartwatch) are highlighted in green.

Normal stock levels remain uncolored.

Question 9 : You are given a dataset : Create a drop-down list of product names in a separate cell using Data Validation, and write a formula in the adjacent cell so that when a product is selected from the drop-down, its corresponding price is automatically displayed.

Answer :

Product Drop-Down with Price Lookup

We are given a dataset of products with their prices. The task is to create a drop-down list of product names in one cell, and display the corresponding price in the adjacent cell automatically.

1. Create the Dataset

Example table (assume it is in range A2:B11):

Product Name	Price
Laptop	55000
Smartphone	30000
Tablet	20000
Headphones	2500
Keyboard	1500
Mouse	800
Monitor	12000
Printer	7000
Router	3500
Smartwatch	9000

2. Create Drop-Down List

Select a cell (e.g., D2) where you want the drop-down.

Go to Data → Data Validation.

In Allow, choose List.

In Source, select the product names range (A2:A11).

Now cell D2 contains a drop-down list of product names.

3. Write Formula to Display Price

In the adjacent cell (e.g., E2), enter the formula:

=VLOOKUP(D2, A2:B11, 2, FALSE)

D2 → Selected product from drop-down.

A2:B11 → Table range containing product names and prices.

2 → Column number of the price in the table.

FALSE → Ensures exact match.

4. Outcome

When a product is selected from the drop-down in D2, its corresponding price is automatically displayed in E2.

Example: Selecting Tablet shows 20000 in the adjacent cell.

Final Deliverable

This setup provides an interactive way to quickly check product prices:

Drop-down list for product selection.

Formula (VLOOKUP) for automatic price display.

Useful for inventory management, billing, or product catalogs.

Question 10 : Case Scenario :- You are a data analyst working for a retail firm. You've received a sales dataset that includes customer transactions from various regions and product categories. Your job is to analyze this data, clean inconsistencies, extract insights, and build visual reports. Task : 1. Apply data validation in the Units Sold column to ensure only values between 1 and 20 are allowed. 2. Highlight rows where Profit > ₹5000 3. Highlight any phone number that is not exactly 10 digits 4. Find the Revenue generated by each product and also label according to revenue generated as "High", "Medium", "Low". 5. Find the profit percentage up to two decimal places. 6. Show Total Revenue and Total Profit by Region. 7. Show Average Units Sold per Product Category.

Answer :

1. Apply Data Validation in Units Sold Column

Select the Units Sold column.

Go to Data → Data Validation.

In Allow, choose Whole Number.

Set Minimum = 1 and Maximum = 20.

This ensures only values between 1 and 20 are allowed.

2. Highlight Rows where Profit > ₹5000

Select the dataset.

Go to Conditional Formatting → New Rule → Use a formula.

Enter formula (assuming Profit is in column F):

=$F2>5000

Set formatting to fill color (e.g., light green).

Entire row will be highlighted if Profit > ₹5000.

3. Highlight Phone Numbers Not Exactly 10 Digits

Select the Phone Number column.

Conditional Formatting → New Rule → Use a formula.

Formula (assuming Phone Number is in column G):

=LEN(G2)<>10

Apply formatting (e.g., red fill).

Any number not 10 digits will be highlighted.

4. Find Revenue Generated by Each Product and Label

Formula for Revenue (assuming Units Sold in column C and Price in column D):

=C2*D2

To categorize Revenue:

=IF(E2>50000,"High",IF(E2>=20000,"Medium","Low"))

This labels each product as High, Medium, or Low revenue.

5. Find Profit Percentage (Two Decimal Places)

Formula (assuming Profit in column F and Revenue in column E):

=ROUND((F2/E2)*100,2)

Displays profit percentage up to two decimals.

6. Show Total Revenue and Total Profit by Region

Insert a Pivot Table.

Rows → Region.

Values → Sum of Revenue, Sum of Profit.

This shows total revenue and profit per region.

7. Show Average Units Sold per Product Category

Insert another Pivot Table.

Rows → Product Category.

Values → Average of Units Sold.

This displays average units sold per category.

Final Deliverable

The cleaned and analyzed dataset provides:

Validated Units Sold entries (1–20 only).

Conditional formatting to highlight high profits and invalid phone numbers.

Revenue categorization into High, Medium, Low.

Profit percentage calculations with precision.

Pivot Table reports showing total revenue/profit by region and average units sold per category.

This ensures the retail firm has a clear, interactive, and accurate view of sales performance across regions and product categories.